Optimal Maximum-Likelihood Tie-Breaking and a Reliability–Sensitivity Identity for Decoded Quantum Interferometry

Yuwen Huang, Wai Ho Mow, and John C. S. Lui

***Abstract*—A deterministic syndrome decoder used coherently in decoded quantum interferometry (DQI) determines both its accepted probability and which accepted errors interfere through one-bit pairs. We optimize the latter quantity while holding classical maximum-likelihood (ML) performance fixed. The model is the phase-averaged separable binary DQI accepted-pair functional, with independent probabilities $0 < p_i < 1/2$, positive coordinate rewards, and deterministic exact-ML syndrome sections. We solve this complete tie class. The optimal rule minimizes weighted log-likelihood, maximizes an explicit reward-weighted downward-edge potential within every ML tie, and resolves the remaining tie by a degree-compatible monomial order. Its representatives are downward closed, and an exact two-term defect quantifies every competing ML section.**

The same optimizer yields a reliability–sensitivity identity: its optimal interference mass equals an ideal product-state term plus a directional derivative of the classical ML success probability. In the identical-bit, uniform-reward case, integration gives a finite-blocklength area law that makes the complete optimal DQI-deficit curve and the ML reliability curve determine one another. At every syndrome rate r, there exist full-rank linear syndrome-map sequences whose optimal exact-ML score has the explicit Shannon-threshold limit $1/2 + \min\{p, q_r\}\sqrt{(1-p)/p}$ for the optimal score, where $h_2(q_r) = r$. The canonical section is also coefficientwise optimal for every nonnegative radial amplitude profile. Finally, a sharpened ambiguity-mass bound identifies a no-gain regime. Complementary constructions give a positive-rate separation with polynomial-time exact ML and a connected, coordinate-indecomposable linear-time witness without weight-two codewords. Independent exhaustive certificates verify the identities, constants, and stated scope.

***Index Terms*—Decoded quantum interferometry, maximum-likelihood decoding, decoder ties, reliability functions, coset leaders, product measures.**

I. Introduction

MAXIMUM-LIKELIHOOD decoding treats each received syndrome separately. If several representatives have the same largest likelihood, any deterministic tie rule gives the same frame-success probability. This observation makes tie-breaking largely invisible to conventional block-error analysis; indeed, decoder ties do not change the error exponent of a code sequence on the binary symmetric channel (BSC) [2], [3].

Decoded quantum interferometry (DQI) changes the performance functional. DQI uses a reversible classical decoder inside a quantum circuit to reduce a structured optimization problem to decoding [1]. After postselection, the output quality depends both on the probability mass accepted by the decoder and on one-bit pairs of accepted errors. Two decoders can therefore have the same likelihood in every syndrome and still prepare different states. The distinction is operational: changing a tie at one syndrome can destroy a one-bit pair with a neighboring syndrome even though neither decoder incurs an additional classical frame error.

This paper asks a narrow coding-theoretic question.

> *Among deterministic ML syndrome decoders with identical conventional performance, which tie rule maximizes the DQI accepted-pair objective?*

The question differs from designing a decoder for bit-error probability or another classical loss [4]. It also differs from improving the shell-wise imperfect-decoding bounds in the original DQI analysis and recent extensions [1], [20]. Shell counts describe how much probability mass a decoder accepts at each Hamming weight. The exact DQI term also records how those accepted errors are joined by edges of the Hamming cube. The latter is a two-point statistic. The main result below nevertheless shows that its unrestricted ML optimum has a universal canonical form.

A. Contributions and scope

Within the phase-averaged separable binary accepted-pair model, the paper makes four principal contributions.

1) We solve the complete class of deterministic exact-ML syndrome sections for independent binary product priors $0 < p_i < 1/2$ and positive coordinate rewards. The optimal syndrome-local rule minimizes weighted log-likelihood, maximizes an explicit reward-weighted downward-edge potential, and then uses a fixed degree-compatible multiplicative monomial order. Its image is downward closed. An exact gap identity separates fiberwise potential loss from missing boundary edges and characterizes every optimizer.
2) We connect the optimum exactly to classical decoding reliability. The optimal accepted-edge mass is an ideal product-state term plus a one-sided directional derivative of $Z_{\mathrm{ML}}(\boldsymbol{p})$. In the radial case this becomes a differential identity and a finite-blocklength area law for $-m^{-1}\log Z_{\mathrm{ML}}(p)$. Product-measure derivative formulas are classical [5], [6]; the new point is that DQI-optimal ML tie-breaking realizes the active directional derivative and converts it into an exact interference score. As a concrete coding consequence, we prove the existence of

Manuscript draft, August 2026. Author affiliations and funding acknowledgments will be inserted before submission.

full-rank linear syndrome-map sequences at every fixed syndrome rate and derive for those sequences a sharp Shannon threshold, correct-decoding exponent, limiting optimal DQI score, and integrated DQI deficit.

3) In the identical-bit, uniform-reward setting, one canonical section is coefficientwise optimal in every Hamming layer and therefore maximizes every nonnegative radial accepted-edge mass simultaneously. The monotone structure of suitably ordered coset leaders is classical [9], [10], [12], [13]; our result is its DQI coefficientwise optimality, closed-form optimum, and exact defect.
4) We prove a stability bound in terms of the total selected ML mass on ambiguous syndromes. If this mass vanishes relative to acceptance, all deterministic ML tie rules have asymptotically identical DQI scores. A quotient lift and a coordinate-indecomposable fixed-distance family show that a nonvanishing separation is possible when an ambiguity budget remains. A logarithmic-block construction further gives positive rate, nonvanishing acceptance, and polynomial-time exact ML. In all cases the compared rules have identical ML likelihood, acceptance, Hamming-layer profile, and matched asymptotic decoding cost.

The accepted-pair expression itself comes from the imperfect-decoder DQI analysis [1]; coordinate-weighted versions appear in multivariate DQI [22]. Weighted-Hamming ML decoding, canonical coset-leader down-sets, Maximum Section, and DE-NFG theory are also prior tools [11], [14], [18]. We use them without claiming their invention. Secondary sections record a paired estimator, a restricted compatible-section formulation, and a real-PSD DE-NFG realization for offline evaluation.

The results do not establish quantum supremacy, a universal gain over all ML decoders, or an asymptotic gain for generic LDPC ensembles. Such claims depend on the optimization instance and on the full reversible cost. Recent work likewise emphasizes that DQI advantage is structure dependent [21], [23].

II. Decoder Sections and the DQI Pair Functional

Unless a base is displayed explicitly, $\log$ denotes the natural logarithm.

Let $H \in \mathbb{F}_2^{\rho\times m}$ have full row rank, let $C = \ker H$, and let $\mathcal{S} = \operatorname{im} H = \mathbb{F}_2^{\rho}$. A deterministic syndrome decoder is a section

$$D : \mathcal{S} \longrightarrow \mathbb{F}_2^m, \qquad HD(s) = s. \tag{1}$$

Its image $T_D = D(\mathcal{S})$ contains one representative from every coset of C. We interchangeably call D a decoder and T_D a transversal.

Write $\mathbf{e}_i$ for the ith unit vector and e_i for coordinate i of e. Let $\{a_e : e \in \mathbb{F}_2^m\}$ be nonnegative real amplitudes normalized by $\sum_e a_e^2 = 1$. Let coordinate i have reward $\gamma_i > 0$ and put $\Gamma = \sum_i \gamma_i$. Define the accepted norm and weighted undirected accepted-edge mass

$$Z_{D,a} = \sum_{e\in T_D} a_e^2, \tag{2}$$

$$\mathcal{E}_{D,a,\gamma} = \sum_{e\in T_D} \sum_{\substack{i:e_i=0\\ e+\mathbf{e}_i\in T_D}} \gamma_i a_e a_{e+\mathbf{e}_i}. \tag{3}$$

Every Hamming-cube edge is counted once.

Theorem 1 (Generalized accepted-pair identity)**.** *For the phase-averaged binary DQI objective,*

$$\mathcal{J}_{D,a,\gamma} = \frac{1}{2} + \frac{\mathcal{E}_{D,a,\gamma}}{\Gamma Z_{D,a}}, \tag{4}$$

$$R_{D,a,\gamma} := Z_{D,a}\left(\mathcal{J}_{D,a,\gamma} - \frac{1}{2}\right) = \frac{\mathcal{E}_{D,a,\gamma}}{\Gamma}. \tag{5}$$

Proof. Jordan *et al.* derive the deterministic imperfect-decoder pair expression for shell amplitudes, and Bu *et al.* give its weighted block-objective extension [1, Supplementary Secs. 10.3–10.4] [22]. For completeness, the phase-averaged functional is the expectation of the following normalized accepted-state observable:

$$|\phi_{D,a}\rangle = \frac{1}{\sqrt{Z_{D,a}}} \sum_{e\in T_D} a_e|e\rangle, \qquad O_\gamma = \frac{1}{2}I + \frac{1}{2\Gamma}\sum_i \gamma_i X_i. \tag{6}$$

Expanding $\langle\phi_{D,a}|O_\gamma|\phi_{D,a}\rangle$ counts both orientations of every accepted i-edge through X_i; the factor $1/2$ leaves each undirected edge once. This proves (4), and multiplication by $Z_{D,a}$ proves (5). □

The theorem is a direct arbitrary-amplitude, separable-reward extension of the cited DQI identities; the results below optimize the decoder within this functional. The rewards γ_i cover the separable coordinate-weighted binary case motivated by multivariate DQI [22]; they do not cover its complete polynomial family. A radial profile has $a_e = a_{\mathrm{wt}(e)}$ and normalization $\sum_k \binom{m}{k} a_k^2 = 1$. For the Bernoulli radial profile, fix $0 < p < 1/2$, set $\gamma_i = 1$, and write

$$a_k = p^{k/2}(1-p)^{(m-k)/2}, \qquad q = \frac{p}{1-p}. \tag{7}$$

The product-amplitude specialization is operationally elementary: for independent probabilities p_i, the input state is

$$|\psi_{\boldsymbol{p}}\rangle = \bigotimes_{i=1}^{m}\left(\sqrt{1-p_i}\,|0\rangle + \sqrt{p_i}\,|1\rangle\right). \tag{8}$$

It is prepared by one single-qubit rotation per coordinate, followed by the same coherent evaluation and uncomputation of the deterministic decoder used in DQI. Thus the product-prior results optimize an efficiently preparable separable specialization of the phase-averaged binary functional rather than an abstract amplitude oracle. We do not assert that every finite-degree shell-polynomial state in the original DQI construction has this product form. Define

$$W_D(q) = \sum_{e\in T_D} q^{\mathrm{wt}(e)}, \tag{9}$$

$$P_D(q) = \sum_{\substack{\{e,e'\}\subseteq T_D\\ d_H(e,e')=1}} q^{\min\{\mathrm{wt}(e),\mathrm{wt}(e')\}}. \tag{10}$$

Then Theorem 1 specializes to

$$Z_D = (1-p)^m W_D(q), \tag{11}$$

$$\mathcal{J}_D = \frac{1}{2} + \frac{\sqrt{q}}{m}\frac{P_D(q)}{W_D(q)}, \tag{12}$$

$$R_D = (1-p)^m \frac{\sqrt{q}}{m} P_D(q). \tag{13}$$

The decomposition is central. Conventional ML decoding maximizes one vertex weight independently in every coset. Equation (10) couples adjacent cosets. Consequently, equal W_D does not imply equal P_D.

A. A paired-sampling identity

The pair term can be estimated without enumerating the transversal. Draw a coordinate I uniformly from $\{1,\dots,m\}$ and draw the other $m-1$ bits i.i.d. from $\mathrm{Bernoulli}(p)$. Let $E^{(0)}$ and $E^{(1)}$ be the two errors obtained by fixing bit I to zero and one. Define $A_D(e) = \mathbf{1}\{D(He) = e\}$.

Proposition 2 (Unbiased pair estimator)**.** *Put*

$$\Xi_D := \Pr\{A_D(E^{(0)}) = 1,\ A_D(E^{(1)}) = 1\}. \tag{14}$$

Then

$$R_D = \frac{(1-p)^m\sqrt{q}}{m} P_D(q) = \sqrt{p(1-p)}\,\Xi_D. \tag{15}$$

Thus two deterministic decoder calls give a bounded unbiased sample of R_D.

Proof. For a fixed coordinate and fixed values of the other bits, the sampling probability of the common $m-1$-bit pattern is exactly the product prior with the varying bit removed. Multiplication by $\sqrt{p(1-p)}$ restores the geometric mean of the two endpoint probabilities. Averaging over I counts every accepted undirected edge once and divides by m. □

Hoeffding's inequality gives additive accuracy ε with $O(\varepsilon^{-2}\log(1/\delta))$ paired samples at confidence $1-\delta$, independent of blocklength. Estimating the conditional score additionally requires stable division by Z_D; this is well conditioned when Z_D is bounded below. Equation (15) is the matched baseline for any approximate DE-NFG or Bethe estimator.

III. Universal Optima over ML Tie Rules

A. Asymmetric product priors

First consider independent, nonidentical error bits with $0 < p_i < 1/2$ and

$$\pi(e) = \prod_{i=1}^{m} p_i^{e_i}(1-p_i)^{1-e_i}. \tag{16}$$

Use the product amplitudes $a_e = \sqrt{\pi(e)}$ and retain arbitrary positive coordinate rewards γ_i from Section II. In this subsection, write $\mathcal{E}_{\gamma,T} := \mathcal{E}_{T,\sqrt{\pi},\gamma}$ and $Z_T := Z_{T,\sqrt{\pi}}$. Define

$$\lambda_i = \log\frac{1-p_i}{p_i}, \qquad r_i = e^{\lambda_i/2} = \sqrt{\frac{1-p_i}{p_i}}, \tag{17}$$

and, for a binary word e,

$$L(e) = \sum_i \lambda_i e_i, \qquad G_\gamma(e) = \sum_{i:e_i=1} \gamma_i r_i. \tag{18}$$

Exact ML decoding minimizes $L(e)$ within each syndrome class. The second quantity is the total reward-weighted amplitude of all downward incidences charged to e.

Theorem 3 (Optimal ML tie rule for product priors and coordinate rewards)**.** *For every product prior* (16) *and every $\gamma_i > 0$, apply the ordered rule*

$$\begin{aligned} &\textit{primary: minimize } L(e),\\ &\textit{secondary: maximize } G_\gamma(e),\\ &\textit{tertiary: take the least word in a fixed degree-compatible}\\ &\qquad\qquad \textit{multiplicative monomial order}. \end{aligned} \tag{19}$$

Its image $T_\star$ is downward closed and globally maximizes the DQI accepted-edge mass over all deterministic ML sections.

More precisely, let e_s be the representative chosen by any ML section T, let $e_s^\star$ be the choice in (19)*, and let $\pi_s = \pi(e_s) = \pi(e_s^\star)$. Define*

$$B_\downarrow(T) = \sum_s \pi_s \sum_{i:e_{s,i}=1} \gamma_i r_i \mathbf{1}\{e_s + \mathbf{e}_i \notin T\}. \tag{20}$$

Then the exact accepted-edge mass and exact optimality gap are

$$\mathcal{E}_{\gamma,T} = \sum_s \pi_s G_\gamma(e_s) - B_\downarrow(T), \tag{21}$$

$$\mathcal{E}_{\gamma,T_\star} - \mathcal{E}_{\gamma,T} = \sum_s \pi_s\left[G_\gamma(e_s^\star) - G_\gamma(e_s)\right] + B_\downarrow(T). \tag{22}$$

Both terms in (22) *are nonnegative. Consequently, an ML section is optimal if and only if it maximizes G_γ in every ML fiber and its image is downward closed. All ML sections have the same acceptance $Z_{\mathrm{ML}} = \sum_s \pi_s$, so $T_\star$ also maximizes the conditional score $1/2 + \mathcal{E}_{\gamma,T}/(\Gamma Z_{\mathrm{ML}})$ and its acceptance-weighted per-attempt excess. Scaling all γ_i by a common constant does not change the optimizer.*

Proof. If $e_i = 1$, then

$$\gamma_i\sqrt{\pi(e)\pi(e+\mathbf{e}_i)} = \pi(e)\gamma_i r_i. \tag{23}$$

Charging every accepted edge to its endpoint containing the one in its varying coordinate proves (21). The rule in (19) maximizes the first term independently in every ML fiber.

It remains to prove downward closure. Let e be selected with $e_i = 1$ and put $f = e + \mathbf{e}_i$. The word f is ML in its coset. Indeed, if a coset mate v had $L(v) < L(f)$, then $v + \mathbf{e}_i$ would lie in the coset of e and have cost at most $L(v) + \lambda_i < L(e)$. Suppose that the rule selects $f' \neq f$ in the coset of f. Both have the same cost. Coordinate i cannot belong to $\mathrm{supp}(f')$, because then $L(f' + \mathbf{e}_i) = L(e) - 2\lambda_i$. Hence $f' + \mathbf{e}_i$ is an ML candidate in the coset of e, and

$$G_\gamma(f' + \mathbf{e}_i) - G_\gamma(e) = G_\gamma(f') - G_\gamma(f). \tag{24}$$

If $G_\gamma(f') > G_\gamma(f)$, this contradicts the secondary choice of e. If the two values agree, adjoining the same absent coordinate preserves the multiplicative monomial order and contradicts the tertiary choice. Thus f is selected. Downward closure gives $B_\downarrow(T_\star) = 0$; subtracting (21) for T from that for $T_\star$ proves (22). □

The rule remains syndrome-local. An exact weighted-ML list decoder first retains all minimum-L candidates, then takes the largest G_γ, and finally uses the fixed monomial-order key. A trellis or junction-tree solver can carry $(L, -G_\gamma, \text{key})$ as its ordered path metric. This resolves tie design; it does not make exact ML decoding polynomial for arbitrary codes. The asymmetric theorem is most relevant when reliabilities repeat or are grouped or quantized; with generic incommensurate likelihoods, most fibers have no ML tie and the secondary rule is vacuous.

The same rule has a second interpretation that links the DQI optimum directly to a standard classical reliability function. Define the optimal ML success probability

$$Z_{\mathrm{ML}}(\boldsymbol{p}) := \sum_{s\in\mathcal{S}} \max_{e:He=s} \pi_{\boldsymbol{p}}(e). \tag{25}$$

For a direction $\boldsymbol{v}$, let $D^+_{\boldsymbol{v}}$ denote the right directional derivative.

Theorem 4 (Reliability-sensitivity identity)**.** *Fix $0 < p_i < 1/2$ and $\gamma_i > 0$, and define*

$$v_i = \gamma_i \sqrt{p_i}(1-p_i)^{3/2}, \tag{26}$$

$$b_\gamma(\boldsymbol{p}) = \sum_{i=1}^{m} \gamma_i \sqrt{p_i(1-p_i)}. \tag{27}$$

Then the right directional derivative of (25) *exists and*

$$\max_{T\ \mathrm{ML}} \mathcal{E}_{\gamma,T} = D^+_{\boldsymbol{v}} Z_{\mathrm{ML}}(\boldsymbol{p}) + b_\gamma(\boldsymbol{p}) Z_{\mathrm{ML}}(\boldsymbol{p}), \tag{28}$$

$$\max_{T\ \mathrm{ML}} \mathcal{J}_{\gamma,T} = \frac{1}{2} + \frac{b_\gamma(\boldsymbol{p})}{\Gamma} + \frac{1}{\Gamma} D^+_{\boldsymbol{v}} \log Z_{\mathrm{ML}}(\boldsymbol{p}). \tag{29}$$

The directional derivative is nonpositive. Hence the ideal product-state bias b_γ/Γ is an upper bound, and its exact loss is the directional sensitivity of the classical ML success probability.

Proof. Along $\boldsymbol{p} + t\boldsymbol{v}$,

$$\frac{d}{dt} \log \pi_{\boldsymbol{p}+t\boldsymbol{v}}(e)\bigg|_{t=0} = G_\gamma(e) - b_\gamma(\boldsymbol{p}). \tag{30}$$

The maximum in each syndrome fiber is over a finite set. The right derivative of that maximum is therefore the largest derivative among its active ML candidates. The secondary rule in Theorem 3 selects exactly that candidate. Summing (30) over syndrome fibers gives (28); division by the common acceptance and by Γ gives (29).

The selected set is downward closed. The probability of any fixed downward- closed subset of the Boolean cube is nonincreasing when any p_i increases, by the standard monotone coupling of product Bernoulli measures. The active section selected above realizes the right derivative of Z_{ML}, so $D^+_{\boldsymbol{v}} Z_{\mathrm{ML}} \le 0$. □

For the identical-bit Bernoulli profile, the sensitivity identity has an exact integral form. Let $\mathcal{J}^\star_{\mathrm{ML}}(p)$ denote the optimum over deterministic ML sections and define its deficit from the full-cube bias by

$$\mathcal{L}_{\mathrm{DQI}}(p) := \frac{1}{2} + \sqrt{p(1-p)} - \mathcal{J}^\star_{\mathrm{ML}}(p) \ge 0. \tag{31}$$

Corollary 5 (Radial derivative and area identities)**.** *For $0 < p < 1/2$,*

$$\mathcal{J}^\star_{\mathrm{ML}}(p) = \frac{1}{2} + \sqrt{p(1-p)} + \frac{\sqrt{p}(1-p)^{3/2}}{m} \frac{d}{dp} \log Z_{\mathrm{ML}}(p), \tag{32}$$

$$-\frac{1}{m} \log Z_{\mathrm{ML}}(p) = \int_0^p \frac{\mathcal{L}_{\mathrm{DQI}}(u)}{\sqrt{u}(1-u)^{3/2}}\, du. \tag{33}$$

Thus the complete optimal DQI deficit curve and the classical ML reliability curve determine one another.

Proof. When all $p_i = p$ and all $\gamma_i = 1$, the direction in (26) is constant across coordinates. Hence $D^+_{\boldsymbol{v}} \log Z_{\mathrm{ML}} = \sqrt{p}(1-p)^{3/2}(d/dp) \log Z_{\mathrm{ML}}$, while $b_\gamma = m\sqrt{p(1-p)}$ and $\Gamma = m$. Equation (29) therefore gives (32). Rearranging and integrating from zero, where $Z_{\mathrm{ML}}(0) = 1$, proves (33). □

This identity is distinct from the EXIT/GEXIT area theorem for channel decoding [7]. EXIT/GEXIT integrates an extrinsic-information quantity and connects iterative and a posteriori decoding thresholds. Here the integrand is the phase-averaged DQI accepted-pair deficit, and the identity holds because the DQI-optimal ML tie rule realizes the active derivative of the exact correct-decoding probability. No belief-propagation approximation or extrinsic-information interpretation enters the statement.

Define the normalized classical ML reliability exponent

$$I_m(p) := -\frac{1}{m} \log Z_{\mathrm{ML},m}(p). \tag{34}$$

Corollary 6 (Asymptotic area law)**.** *For every code sequence and $0 \le a < b < 1/2$,*

$$\int_a^b \frac{\mathcal{L}_{\mathrm{DQI},m}(u)}{\sqrt{u}(1-u)^{3/2}}\, du = I_m(b) - I_m(a). \tag{35}$$

Hence convergence of I_m at the two endpoints determines the limiting weighted area on $[a,b]$ without any exchange of derivatives and limits. In particular, subexponential ML success loss at p, $Z_{\mathrm{ML},m}(p) = \exp[-o(m)]$, forces the weighted area on $[0,p]$ to vanish; an exponent $Z_{\mathrm{ML},m}(p) = \exp[-mI(p) + o(m)]$ gives limiting area $I(p)$.

Proof. Subtract (33) at a and b. The two asymptotic statements then follow directly and require no pointwise convergence of the deficit. □

B. A Shannon-threshold consequence for linear syndrome maps

The area law becomes explicit for fixed-rate linear source coding. Write

$$h_2(x) = -x \log_2 x - (1-x)\log_2(1-x), \tag{36}$$

$$D_{\mathrm{b}}(x\|p) = x \log \frac{x}{p} + (1-x) \log \frac{1-x}{1-p}. \tag{37}$$

For a syndrome rate $r = \rho/m \in (0,1)$, equivalently a channel-code rate $1-r$, let $q_r \in (0, 1/2)$ be the unique solution of $h_2(q_r) = r$. The exponent below is the Bernoulli specialization of the fixed-rate correct-decoding exponent; attainability by

linear syndrome codes is classical [8]. We include a direct argument because the resulting DQI score law is new.

Theorem 7 (Linear-syndrome Shannon threshold)**.** *For every $r \in (0,1)$, with $\rho_m = \lfloor rm \rfloor$, there exists a sequence of full-row-rank binary maps $H_m \in \mathbb{F}_2^{\rho_m \times m}$ such that, simultaneously for every fixed $0 < p < 1/2$,*

$$\lim_{m\to\infty} I_m(p) = I_r(p) := \begin{cases} 0, & p \le q_r, \\ D_{\mathrm{b}}(q_r \| p), & p > q_r. \end{cases} \tag{38}$$

For the canonical DQI-optimal deterministic exact-ML section with uniform coordinate rewards,

$$\lim_{m\to\infty} \mathcal{J}^{\star}_{\mathrm{ML},m}(p) = \frac{1}{2} + \min\{p, q_r\}\sqrt{\frac{1-p}{p}}. \tag{39}$$

Consequently,

$$\lim_{m\to\infty} \int_0^p \frac{\mathcal{L}_{\mathrm{DQI},m}(u)}{\sqrt{u}(1-u)^{3/2}}\, du = I_r(p). \tag{40}$$

Thus the classical source-coding threshold $r = h_2(p)$ is also the point at which the limiting optimal DQI score leaves its full-cube value.

Proof. Every syndrome transversal contains exactly 2^{ρ_m} words. Since $p < 1/2$, Bernoulli word probabilities decrease with Hamming weight. Hence the success mass of any transversal is at most the mass of the 2^{ρ_m} most probable words. Hamming-ball asymptotics and the method of types give the universal lower bound

$$\liminf_{m\to\infty} I_m(p) \ge I_r(p). \tag{41}$$

For achievability, draw a $\rho_m \times m$ binary matrix with independent uniform entries, where $\rho_m = \lfloor rm \rfloor$. Put $\epsilon_m = m^{-1/3}$ and $a_m = \lfloor m(q_r - \epsilon_m) \rfloor$. For an error e of weight w, a union bound gives

$$\begin{aligned} \Pr_H\{\exists e' \neq e :\ \mathrm{wt}(e') \le w,\ He' = He\} \\ \le 2^{-\rho_m} \sum_{j=0}^{w} \binom{m}{j}. \end{aligned} \tag{42}$$

Uniformly for $w \le a_m$, the right side is at most $\exp[-cm^{2/3}]$ for some $c = c(r) > 0$ and all sufficiently large m. Markov's inequality at level $\exp[-cm^{2/3}/2]$, followed by a union bound over the $m+1$ shells, shows that, with probability tending to one, the fraction of colliding words in every shell $w \le a_m$ is $o(1)$. A random rectangular binary matrix is also full row rank with probability tending to one. Select one matrix satisfying both properties. Every noncolliding word is the unique ML representative of its syndrome, so its ML success probability obeys

$$Z_{\mathrm{ML},m}(p) \ge (1 - o(1)) \Pr\{\mathrm{Bin}(m,p) \le a_m\}. \tag{43}$$

For $p < q_r$ the right side tends to one. For $p = q_r$ it is $\exp[-o(m)]$, and for $p > q_r$ its exponent tends to $D_{\mathrm{b}}(q_r\|p)$. Together with (41), this proves (38). The shell property is independent of p, so the selected sequence works simultaneously for every fixed p. At the boundary $p = q_r$, the conclusion is only a zero normalized exponent; it does not assert that $Z_{\mathrm{ML},m}(q_r)$ tends to one.

Let T_m be the canonical downward-closed ML section and define its accepted conditional relative weight

$$\bar{w}_m(p) := \frac{1}{mZ_{\mathrm{ML},m}(p)} \sum_{e\in T_m} \pi_p(e)\, \mathrm{wt}(e). \tag{44}$$

If $p < q_r$, (43) gives $Z_{\mathrm{ML},m}(p) \to 1$; because $0 \le \mathrm{wt}(e)/m \le 1$, conditioning on acceptance then gives $\bar{w}_m(p) \to p$. Let $p \ge q_r$ and fix a sufficiently small $\delta > 0$. The accepted mass below $(q_r - \delta)m$ is bounded by the corresponding Bernoulli lower tail, whose exponent is strictly larger than $I_r(p)$. The accepted mass above $(q_r+\delta)m$ is at most 2^{ρ_m} times the largest probability of a word in that range; its exponent is also strictly larger than $I_r(p)$. Dividing these two bounds by $Z_{\mathrm{ML},m}(p) = \exp[-mI_r(p)+o(m)]$ shows that both conditional tails vanish. Since $\delta > 0$ is arbitrary, $\bar{w}_m(p) \to q_r$.

Every one-bit edge of a downward-closed section can be charged to its upper endpoint. Therefore the exact optimal score is

$$\mathcal{J}^{\star}_{\mathrm{ML},m}(p) = \frac{1}{2} + \sqrt{\frac{1-p}{p}}\, \bar{w}_m(p), \tag{45}$$

which proves (39). Finally, (40) follows from Corollary 6 and (38). □

The limiting score in (39) is continuous and strictly increases up to $p = q_r$, then strictly decreases. It therefore has a unique cusp maximum $1/2 + \sqrt{q_r(1-q_r)}$ at the Shannon threshold. Above threshold, the limiting deficit is explicitly $(p-q_r)\sqrt{(1-p)/p}$, whose area-law integrand is $\frac{d}{dp} D_{\mathrm{b}}(q_r\|p)$.

C. A coefficientwise radial companion

Set $\gamma_i = 1$, suppress γ from the notation, and let the amplitudes depend only on Hamming weight. For each syndrome s, let

$$\mathcal{L}_s = \arg\min_{e:He=s} \mathrm{wt}(e), \qquad \ell_s = \mathrm{wt}(e) \quad (e \in \mathcal{L}_s), \tag{46}$$

and let $n_k = |\{s : \ell_s = k\}|$. Every deterministic ML section selects exactly n_k words in layer k. For an ML transversal T, let $N_k(T)$ be the number of accepted Hamming edges between layers $k-1$ and k.

Theorem 8 (Coefficientwise optimal ML section)**.** *For every binary linear code and every deterministic ML section T,*

$$N_k(T) \le kn_k, \qquad 1 \le k \le m. \tag{47}$$

Fix a coordinate order and, within every coset, choose a minimum-weight word; among ties, choose the lexicographically first word with $0 < 1$. The resulting section T_{glex} attains equality simultaneously for every k. Consequently, T_{glex} maximizes $\mathcal{E}_{T,a}$ and $R_{T,a}$ for every nonnegative radial amplitude profile a. It also maximizes the conditional score $\mathcal{J}_{T,a}$ whenever the common ML acceptance $Z_{\mathrm{ML},a}$ is positive.

More precisely, define the layer defect

$$\delta_k(T) := kn_k - N_k(T). \tag{48}$$

Then

$$Z_{\mathrm{ML},a} = \sum_{k=0}^{m} n_k a_k^2, \tag{49}$$

$$\mathcal{E}_{T,a} = \sum_{k=1}^{m} a_{k-1} a_k \, [k n_k - \delta_k(T)]. \tag{50}$$

The optimum has every $\delta_k = 0$. If $a_{k-1}a_k > 0$ for all occupied layers, an ML section is optimal exactly when its selected set is downward closed.

Proof. Every selected word e of weight k has exactly k downward neighbors $e + \mathbf{e}_i$, one for each coordinate i with $e_i = 1$. Charge each accepted edge to its upper endpoint. This gives (47), and the missing downward incidences are exactly $\delta_k(T)$. Every edge between layers $k-1$ and k contributes $a_{k-1}a_k$, which proves (50).

It remains to prove the claimed closure. Graded lexicographic order is a degree-compatible multiplicative monomial order $\prec$ on squarefree monomials. More generally, choose the $\prec$-least binary word in every code coset under any such order. Degree compatibility makes each choice an ML representative. We show that the selected set is downward closed. Let e be selected, let $e_i = 1$, and put $f = e + \mathbf{e}_i$. The word f is a coset leader: a lower-weight word in its coset would, after toggling coordinate i, give a word lighter than e in the coset of e. If f were not selected, the selected word f' in its coset would have the same weight and satisfy $f' \prec f$. Coordinate i cannot lie in the support of f', since $f' + \mathbf{e}_i$ would then be lighter than e in the coset of e. Multiplicativity therefore gives $f' + \mathbf{e}_i \prec f + \mathbf{e}_i = e$, again contradicting the choice of e. Hence every downward neighbor of every selected word is selected. □

The rule is syndrome-local, yet its complete image is globally downward closed. The monotone structure of correctable errors, weight-compatible orders, and Gröbner representations of binary codes are classical [9], [10], [12], [13]. The new conclusion here is that their downward-closed canonical section is coefficientwise optimal for the DQI pair functional. Constructing the complete coset-leader table can still be exponential in the code codimension. The theorem gives the exact global target; compact realizations require additional code structure.

For the Bernoulli profile, put

$$W_{\mathrm{ML}}(q) = \sum_{k=0}^{m} n_k q^k. \tag{51}$$

Theorem 8 yields the closed form

$$\max_{D \ \mathrm{ML}} P_D(q) = W'_{\mathrm{ML}}(q), \tag{52}$$

$$\max_{D \ \mathrm{ML}} \mathcal{J}_D = \frac{1}{2} + \frac{\sqrt{q}}{m} \frac{W'_{\mathrm{ML}}(q)}{W_{\mathrm{ML}}(q)}. \tag{53}$$

For any ML section T, the exact polynomial defect is

$$P_T(q) = W'_{\mathrm{ML}}(q) - \Delta_{\downarrow}(T; q), \tag{54}$$

$$\Delta_{\downarrow}(T; q) = \sum_{e \in T} \sum_{i : e_i = 1} q^{\mathrm{wt}(e)-1} \mathbf{1}\{e + \mathbf{e}_i \notin T\}. \tag{55}$$

Thus the same canonical section is optimal for every $0 < p < 1/2$, without retuning the tie rule.

IV. ML Candidate Compatibility and Restricted Tie Design

Let $h_i = H\mathbf{e}_i$ denote column i of H. For each syndrome s, use the complete minimum-weight list $\mathcal{L}_s$ from (46). Every deterministic ML decoder is a selection $D(s) \in \mathcal{L}_s$.

Theorem 9 (Exact compatible-section objective for ML ties)**.** *Every deterministic ML selection has the same denominator*

$$W_{\mathrm{ML}}(q) = \sum_s q^{\ell_s}, \qquad Z_{\mathrm{ML}} = (1-p)^m W_{\mathrm{ML}}(q). \tag{56}$$

Its accepted-edge polynomial is

$$P(D; q) = \frac{1}{2} \sum_{s \in \mathcal{S}} \sum_{i=1}^{m} q^{\min\{\ell_s, \ell_{s+h_i}\}} \mathbf{1}\{D(s+h_i) = D(s) + \mathbf{e}_i\}. \tag{57}$$

Thus every ML tie rule has an exact finite weighted compatible-section representation.

Proof. All candidates in $\mathcal{L}_s$ have weight ℓ_s, which proves (56). Representatives at syndromes s and $s+h_i$ form a physical one-bit pair in coordinate i precisely when the indicator in (57) equals one. The factor $1/2$ removes the two orientations of a syndrome edge. Repeated columns of H remain distinct physical coordinates and are retained in the inner sum. Since the denominator is fixed, all three optimization criteria have the same maximizers. □

An equivalent graph construction makes the optimization precise. Place the list $\mathcal{L}_s$ above each syndrome vertex s. Connect $e \in \mathcal{L}_s$ to $e + \mathbf{e}_i \in \mathcal{L}_{s+h_i}$ with the weight in (57). A decoder selects one lifted vertex above each base vertex, and its reward is the weight of compatible selected pairs. For arbitrary ML lists, an edge may induce only a partial matching, so this is a partial-permutation label problem rather than a covering graph in general. When every edge induces a full bijection on a common label set, it is Maximum Section of a covering graph. The relation of that subclass to Unique Games and Max-2Lin is known [14]. The structure added here is the code-induced quotient, the Bernoulli radial weights, and the DQI pair functional. Theorem 8 already solves the unrestricted ML optimization. The candidate graph remains useful when an implementation restricts the available labels or when a larger, possibly non-ML, candidate set is optimized.

If every list contains two labels and every active edge gives a full matching, label the candidates by $b_s \in \mathbb{F}_2$. Compatibility then has the form

$$b_t = b_s + \delta_{st}. \tag{58}$$

On a tree, labels propagate exactly. On a loopy graph, zero frustration is equivalent to the XOR of δ_{st} vanishing around every cycle. For a label alphabet of size L, exact max-sum on a tree costs $O(\sum_{(s,t)} |\mathcal{L}_s||\mathcal{L}_t|)$, and a graph with N vertices and treewidth κ admits exact junction-tree optimization in $O(NL^{\kappa+1})$ time and $O(NL^{\kappa})$ memory. These are structural algorithms: the explicit syndrome graph has 2^{ρ} vertices, so scalable use requires a compact decoder architecture or an exploitable factorization.

V. A Quotient Lift with an Exact Weighted-Cut Objective

We next construct a class in which a restricted but complete fiber tie has an exact local cut description. The result makes strict losses transparent and gives compact implementations of the canonical optimum.

Let $H_0 \in \mathbb{F}_2^{\rho\times n}$ be a full-row-rank parity-check matrix for a base code $C_0 = \ker H_0$. Write a base word as $y = (a, z)$, where $a \in \mathbb{F}_2$ and $z \in \mathbb{F}_2^{n-1}$. Define the quotient map

$$\begin{aligned} Q &: \mathbb{F}_2^{n+1} \longrightarrow \mathbb{F}_2^n, \\ Qx &= (x_1 + x_2, x_3, \ldots, x_{n+1}), \end{aligned} \tag{59}$$

and set

$$H = H_0 Q. \tag{60}$$

The lifted code $C = \ker H$ contains $g = e_1 + e_2$, and $C/\langle g\rangle \cong C_0$.

Let D_0 be any deterministic base section with image T_0. A binary tie field τ on the selected words $(1, z) \in T_0$ defines

$$L_\tau(a, z) = \begin{cases} (0, 0, z), & a = 0, \\ (\tau(a, z), 1 + \tau(a, z), z), & a = 1. \end{cases} \tag{61}$$

Equivalently, the first two bits are

$$(a\tau(a, z),\ a + a\tau(a, z)). \tag{62}$$

The factors a in (62) are essential: on an $a = 0$ fiber, the prefix 00 is the unique ML choice for $p < 1/2$.

Let $\Gamma_1(T_0)$ be the graph with vertex set

$$V_1 = \{(1, z) \in T_0\}, \tag{63}$$

where two vertices are adjacent if they differ in one z coordinate. Give an edge $e = \{y, y'\}$ weight

$$w_e(q) = q^{\min\{\mathrm{wt}(y),\mathrm{wt}(y')\}}. \tag{64}$$

Theorem 10 (Exact weighted cut over the complete fiber-lift ML class)**.** *For every tie field τ, $L_\tau \circ D_0$ is a valid section of H. If D_0 is exact ML for the base code, every lifted section is exact ML for the lifted code. All tie fields choose a representative of the same weight in every syndrome and satisfy*

$$W_\tau(q) = W_0(q), \qquad Z_\tau(p) = (1 - p)Z_0(p). \tag{65}$$

If $P_0(q)$ is the accepted-edge polynomial of the base section, then

$$P_\tau(q) = P_0(q) - \mathrm{Cut}_\tau(q), \tag{66}$$

$$\mathrm{Cut}_\tau(q) = \sum_{e=\{y,y'\}\in E(\Gamma_1)} w_e(q)\mathbf{1}\{\tau(y) \neq \tau(y')\}. \tag{67}$$

A constant field is therefore globally DQI-optimal over the complete pointwise-ML-equivalent class obtained from the fixed base section D_0. The optimizers are exactly the fields that are constant on each connected component of $\Gamma_1(T_0)$.

Proof. The fiber $Q^{-1}(a, z)$ contains two words that differ by g. For $a = 0$, their prefixes are 00 and 11, so 00 is the unique fiberwise ML choice. For $a = 1$, their prefixes are 01 and 10, which tie. In either case,

$$\max_{Qx=y} \pi_p^{(n+1)}(x) = (1 - p)\pi_p^{(n)}(y). \tag{68}$$

Thus a lifted base ML word is full-code ML and $\mathrm{wt}(L_\tau(y)) = \mathrm{wt}(y)$, proving (65).

It remains to classify physical one-bit pairs. A selected base edge in coordinate a lifts from 00 to either 01 or 10 and remains a physical edge. A z edge in the $a = 0$ slice keeps prefix 00 and also remains. A z edge in the $a = 1$ slice remains precisely when its endpoint labels agree. If they disagree, the prefix changes from 01 to 10 while one z bit also changes, so the lifted distance is three. Conversely, every physical unit vector maps under Q to a base unit vector, so no extra pair is created. This proves (66). Since all cut weights are positive, zero cut is attainable and optimal. It is attained exactly by fields constant on each connected component. □

The scope in Theorem 10 matters. The result is global over all choices of the $01/10$ ambiguity above the fixed base ML section. It does not assert optimality over arbitrary non-ML transversals or over ML base sections outside this fiber-lift class.

A. Cover, gauge, and algorithmic interpretations

The tie graph gives a literal two-sheet covering graph: lift $v \in V_1$ to (v, b), $b \in \mathbb{F}_2$, and connect equal labels above every base edge. A tie field is a section of the cover, and (67) is the cover-edge weight lost by that section. A constant field fixes one sheet globally. With several binary tie coordinates and translation constraints $u_t = u_s + \delta_{st}$, the label group becomes $\mathbb{F}_2^t$ and the problem is Abelian Max-2Lin. The cycle sum of δ_{st} is the cover holonomy. A vertex relabeling gauges all offsets to zero exactly when every cycle sum vanishes.

The aligned lifted decoder has a particularly compact implementation:

$$s \xmapsto{D_0} (a, z) \longmapsto (0, a, z). \tag{69}$$

It calls the base decoder once and performs $O(n)$ reversible bit operations. No exponential syndrome table, fractional-programming loop, or complex message is involved. In a general unfrustrated Abelian lift, choose one root label per component and propagate offsets. For $L = 2^t$ labels, translation constraints reduce exact tree max-sum from $O(|E|L^2)$ to $O(|E|L)$ by keeping the largest and second-largest incoming values.

VI. Reliable Connected Families

The weighted-cut identity would have limited operational force if the common postselection probability vanished. The next result rules out that failure mode.

Corollary 11 (Any reliable base section gives a constant gap)**.** *Let $\{D_{0,n}\}$ be deterministic base sections with success probabilities $Z_{0,n} = 1 - \varepsilon_n$, where $\varepsilon_n \to 0$. In the quotient lift, compare a constant tie field with*

$$\tau_{\mathrm{par}}(1, z) = \bigoplus_{j=1}^{n-1} z_j. \tag{70}$$

Then

$$Z_{\mathrm{const}} = Z_{\mathrm{par}} = (1 - p)Z_{0,n} \longrightarrow 1 - p, \tag{71}$$

$$\mathcal{J}_{\rm const} - \mathcal{J}_{\rm par} \longrightarrow p\sqrt{p(1-p)}, \tag{72}$$

$$R_{\rm const} - R_{\rm par} \longrightarrow [p(1-p)]^{3/2}. \tag{73}$$

Consequently, the expected number of independent postselection attempts $1/Z$ tends to $1/(1-p)$. If amplitude amplification is available, its repetition factor $1/\sqrt{Z}$ tends separately to $1/\sqrt{1-p}$, before the cost of the reversible decoder oracle is charged. No ML property is needed for these limits. If $D_{0,n}$ is exact ML, both lifted sections are exact ML by Theorem 10.

Proof. Equation (71) follows from (65). Parity changes on every z edge, so it cuts every accepted edge of the $a = 1$ slice. For each of the $n-1$ outside coordinates, the complete $a = 1$ edge mass equals $p\sqrt{p(1-p)}$. Relative to the complete slice, a pair is missing only if at least one base endpoint is a decoding failure. With $q = p/(1-p)$, the missing normalized mass in one coordinate is at most $(\sqrt{q} + 1/\sqrt{q})\varepsilon_n$. Summing over the outside coordinates, dividing by the lifted length $n+1$, and using $Z_{0,n} \to 1$ proves (72). Multiplication by the common acceptance proves (73). □

The lift also transfers rate and decoding complexity exactly. A base $[n, k_n]$ code becomes an $[n+1, k_n+1]$ code, so its rate has the same limit. If $D_{0,n}$ runs in time $T(n)$, both lifted sections run in $T(n) + O(n)$ time. Their selected weights and acceptance are identical to each other; the common full-code acceptance is $(1-p)Z_{0,n}$, not $Z_{0,n}$. Hence any reliable positive-rate decoder family gives a positive-rate strict DQI separation. Exact ML status transfers when the base decoder itself is exact ML.

Taking an odd repetition code as the base in Corollary 11 already gives a connected degree-three realization, but the quotient lift contains the weight-two word $e_1 + e_2$. The next family removes this artifact and gives the principal explicit witness.

Theorem 12 (Indecomposable fixed-distance strict-gain family)**.** *Fix even $d \geq 4$ and odd $n \geq d-1$, and put $N = d+n$. On coordinates $(x_1, \ldots, x_d, z_1, \ldots, z_n)$, define*

$$g = (\mathbf{1}^d, 0^n), \qquad h = (e_1, \mathbf{1}^n), \qquad C_{d,n} = \mathrm{span}\{g, h\}. \tag{74}$$

For the inner repetition code $\mathrm{span}\{\mathbf{1}^d\}$, let A_0 and A_1 choose the unique lighter member of every antipodal pair off the middle layer. On a middle-layer pair, A_b chooses the member whose first bit is b. Let

$$T_n = \{z \in \mathbb{F}_2^n : \mathrm{wt}(z) \leq (n-1)/2\}, \tag{75}$$

and define

$$T_{\rm const} = A_0 \times T_n, \tag{76}$$

$$T_{\rm par} = \{(x, z) : z \in T_n, x \in A_{\mathrm{wt}(z) \bmod 2}\}. \tag{77}$$

Both sets are exact ML transversals of $C_{d,n}$ and select a word of the same weight in every coset. Moreover, $T_{\rm const}$ is downward closed and therefore globally DQI-optimal over all deterministic ML sections.

For the Bernoulli profile, put

$$t_d(p) = \frac{1}{2}\binom{d}{d/2}[p(1-p)]^{d/2}, \tag{78}$$

$$\zeta_d(p) = \sum_{j<d/2}\binom{d}{j}p^j(1-p)^{d-j} + t_d(p), \tag{79}$$

$$Z_n^{\rm rep} = \sum_{j=0}^{(n-1)/2}\binom{n}{j}p^j(1-p)^{n-j}, \tag{80}$$

$$E_n^{\rm rep} = \sum_{j=0}^{(n-3)/2}(n-j)\binom{n}{j}p^{j+1/2}(1-p)^{n-j-1/2}. \tag{81}$$

The two sections have common acceptance $\zeta_d Z_n^{\rm rep}$ and exact accepted-edge separation

$$\mathcal{E}_{\rm const} - \mathcal{E}_{\rm par} = t_d E_n^{\rm rep}. \tag{82}$$

For fixed d and $0 < p < 1/2$, as odd $n \to \infty$,

$$\mathcal{J}_{\rm const} - \mathcal{J}_{\rm par} \longrightarrow \frac{t_d(p)}{\zeta_d(p)}\sqrt{p(1-p)}, \tag{83}$$

$$R_{\rm const} - R_{\rm par} \longrightarrow t_d(p)\sqrt{p(1-p)}. \tag{84}$$

The code has minimum distance d, nonzero distinct parity-check columns, a connected check representation of maximum degree three, and linear-time exact ML decoding. It is coordinate-indecomposable.

Proof. Each A_b is a transversal of $\mathrm{span}\{\mathbf{1}^d\}$, and T_n contains one word from every outer antipodal pair. Hence neither selected set contains two words differing by g, h, or $g+h$; its cardinality is 2^{N-2}, so it is a transversal of (74).

Every selected (x, z) satisfies $\mathrm{wt}(x) \leq d/2$ and $\mathrm{wt}(z) \leq (n-1)/2$. Adding g, h, and $g+h$ changes its weight by, respectively,

$$\begin{aligned}\Delta_g &= d - 2\,\mathrm{wt}(x),\\ \Delta_h &= (1-2x_1) + n - 2\,\mathrm{wt}(z),\\ \Delta_{g+h} &= d - 1 - 2(\mathrm{wt}(x) - x_1) + n - 2\,\mathrm{wt}(z).\end{aligned} \tag{85}$$

All three quantities are nonnegative. Thus both sections are exact ML. Their inner representatives have equal weight syndrome by syndrome, which also proves pointwise weight equivalence.

The mass selected by either inner section is ζ_d, and the mass of its selected middle layer is t_d. Along an accepted outer edge, the parity rule switches the inner representative exactly on these middle-layer ties and destroys that one-bit pair; the constant rule preserves it. The inner-edge masses of A_0 and A_1 are equal because every selected middle-layer word has exactly $d/2$ selected downward neighbors. Product factorization then gives (82). Also,

$$\frac{E_n^{\rm rep}}{n} = \sqrt{p(1-p)}\,\Pr\{\mathrm{Bin}(n-1, p) \leq (n-3)/2\}, \tag{86}$$

so $Z_n^{\rm rep} \to 1$ and $E_n^{\rm rep}/n \to \sqrt{p(1-p)}$. Equations (83) and (84) follow.

The sets A_0 and T_n are downward closed, as is their product $T_{\rm const}$. Theorem 8 proves global optimality. The sparse checks

$$x_2 = \cdots = x_d, \qquad z_1 = \cdots = z_n, \qquad x_1 + x_2 + z_1 = 0 \tag{87}$$

have rank $N-2$, maximum degree three, and a connected Tanner graph. The three nonzero codeword weights are d,

$n+1$, and $n+d-1$, so $d_{\min}=d$. This excludes zero and repeated parity-check columns. Every pair among $g, h, g+h$ has intersecting support; a decomposable binary dimension-two code would have a disjoint-support basis. Finally, an arbitrary coset representative has only four codeword shifts. Comparing their weights and applying the stated tie rule gives exact ML decoding in $O(N)$ time. □

The fixed d is substantive: Stirling's formula gives $t_d = \Theta([2\sqrt{p(1-p)}]^d/\sqrt{d})$. Thus any chosen fixed even distance retains a strict constant gap, whereas this particular gap vanishes if d grows. The theorem is an exact-ML structural witness, not a capacity-achieving ensemble or a generic quantum-advantage claim.

A. A positive-rate polynomial-time feasibility family

The preceding indecomposable family has vanishing rate. The following complementary construction shows that positive rate, polynomial-time exact ML, nonvanishing acceptance, and a strict DQI separation can coexist. Its logarithmic block decomposition is deliberately explicit; it is a feasibility result, not a claim about natural LDPC ensembles.

Theorem 13 (Positive-rate polynomial-time exact-ML separation)**.** *Fix $0<p<1/2$, a target channel-code rate $0<R<1-h_2(p)$, and an even integer $d\geq 4$. There exist binary linear codes $\widetilde{C}_N$ and two deterministic exact-ML sections $D_N^{\mathrm{const}}, D_N^{\mathrm{par}}$ with the following properties:*

1) *$\dim(\widetilde{C}_N)/N \to R$ and $d_{\min}(\widetilde{C}_N)=d$ for all sufficiently large N. A full-rank parity-check representation has nonzero distinct columns, a connected Tanner graph, $O(N\log N)$ nonzero entries, and maximum row weight $O(\log N)$.*
2) *Both sections run in $N^{O(1)}$ time and select representatives of the same weight syndrome by syndrome. Their common acceptance satisfies*

$$Z_N^{\mathrm{const}} = Z_N^{\mathrm{par}} \longrightarrow \zeta_d(p) > 0. \tag{88}$$

3) *In the phase-averaged binary accepted-pair model,*

$$\mathcal{J}_N^{\mathrm{const}} - \mathcal{J}_N^{\mathrm{par}} \longrightarrow \frac{t_d(p)}{\zeta_d(p)}\sqrt{p(1-p)} > 0, \tag{89}$$

$$R_N^{\mathrm{const}} - R_N^{\mathrm{par}} \longrightarrow t_d(p)\sqrt{p(1-p)} > 0. \tag{90}$$

For every fixed $\eta>0$, d can be chosen once so that $\zeta_d(p) > 1-\eta$; the two limits in (89)–(90) *then remain positive constants as $N\to\infty$.*

Proof. Choose $\tau\in(p,1/2)$ with $h_2(\tau)<1-R$, and choose $\delta>0$ with $h_2(\delta)<1-R$. For a block length b, draw a binary $\rho_b\times b$ matrix with $\rho_b=\lceil(1-R)b\rceil$. For an error e of weight at most τb, the collision union bound gives

$$\Pr_H\{\exists e'\neq e:\ He'=He,\ \mathrm{wt}(e')\leq \mathrm{wt}(e)\} \leq 2^{-\rho_b}\sum_{j=0}^{\mathrm{wt}(e)}\binom{b}{j}. \tag{91}$$

Together with the Bernoulli tail above τb, this shows that the expected ML block-error probability is at most $\exp(-2\kappa b)$ for some $\kappa=\kappa(p,R)>0$ and all sufficiently large b. Also,

$$\Pr_H\{d_{\min}(\ker H)<\delta b\} \leq 2^{-\rho_b}\sum_{j<\delta b}\binom{b}{j} = \exp[-\Omega(b)]. \tag{92}$$

A random rectangular matrix is full row rank with probability tending to one. Markov's inequality at level $\exp(-\kappa b)$ therefore selects full-row-rank block matrices H_b for which the ML block-error probability is at most $\exp(-\kappa b)$ and the distance bound holds.

Let $L\to\infty$, put $b_L=\lceil A\log L\rceil$ with fixed $A>1/\kappa$, and take the direct sum of L copies of $\ker H_{b_L}$. Its exact ML decoder acts independently on the blocks. The outer success probability $Z_{0,L}$ obeys

$$1-Z_{0,L}\leq L\exp(-\kappa b_L)\longrightarrow 0. \tag{93}$$

Exhaustive ML decoding of one block costs $2^{O(b_L)}$; over all blocks this is polynomial in the outer length $n_L=Lb_L$. The outer rate tends to R and its minimum distance is at least δb_L.

Now take the direct product with the fixed inner repetition code $\mathrm{span}\{\mathbf{1}^d\}$. Use A_0 on every inner antipodal coset for D_N^{const}. For D_N^{par}, replace A_0 by A_1 on the inner middle-layer ties exactly when the selected outer representative has odd Hamming weight. Direct-product ML factorization proves that both are exact ML, and their selected weights agree in every full syndrome. The common acceptance is $\zeta_d Z_{0,L}$.

Let $E_{0,L}$ be the accepted-edge mass of the outer section. Switching the inner orientation destroys precisely the outer accepted edges paired with an inner middle-layer representative. The inner edge masses of A_0 and A_1 are equal, so the finite identity is

$$\mathcal{E}_N^{\mathrm{const}} - \mathcal{E}_N^{\mathrm{par}} = t_d(p)E_{0,L}. \tag{94}$$

The complete n_L-cube has edge mass $n_L\sqrt{p(1-p)}$. Every edge absent from the selected outer set touches a failure word; summing incident edge mass at such words gives

$$0\leq n_L\sqrt{p(1-p)} - E_{0,L} \leq \frac{n_L(1-Z_{0,L})}{\sqrt{p/(1-p)}}. \tag{95}$$

Hence $E_{0,L}/n_L\to\sqrt{p(1-p)}$. Since $N=n_L+d$, (88)–(90) follow.

The full code has dimension $1+L(b_L-\rho_{b_L})$, rate tending to R, and minimum distance $\min\{d,\delta b_L\}=d$. Minimum distance at least four excludes zero and repeated parity-check columns. Start from any full-rank block-diagonal parity-check matrix. In each Tanner component choose one row r_j and replace the selected rows by $r_0, r_0+r_1, r_1+r_2, \ldots$. This is an invertible row operation, connects all components, and raises row weight by at most a factor of two. Because $b_L=O(\log N)$, the resulting representation has the claimed sparsity. Finally, $\zeta_d(p)\to 1$ as even $d\to\infty$ by a binomial-tail bound, while $t_d(p)>0$ for every finite d. □

VII. How Much Can ML Tie-Breaking Change the DQI Score?

The quotient lift introduces persistent weight-two ambiguity. We next show that a nonvanishing selected ambiguous

mass relative to acceptance is necessary for an asymptotic conditional-score gap.

For an ML decoder, the mass of one selected representative from syndrome s is unambiguous because every element of $\mathcal{L}_s$ has the same weight. Define the total selected ML mass on ambiguous syndromes

$$A_{\text{tie}} := \sum_{s:\,|\mathcal{L}_s|>1} \pi_p(e_s), \qquad e_s \in \mathcal{L}_s. \tag{96}$$

For two deterministic ML sections D, D', define the one-sided changed mass

$$\alpha(D, D') := \sum_{s:\,D(s)\neq D'(s)} \pi_p(D(s)). \tag{97}$$

Theorem 14 (DQI diameter of the ML tie class)**.** *Let D, D' be deterministic ML sections of the same code, and let Z be their common acceptance. Then*

$$\alpha(D, D') \le A_{\text{tie}} \le 1 - Z, \tag{98}$$

$$|\mathcal{J}_D - \mathcal{J}_{D'}| \le \frac{\alpha(D, D')}{Z\sqrt{q}} = \frac{\alpha(D, D')}{Z}\sqrt{\frac{1-p}{p}}. \tag{99}$$

Consequently, the diameter of the complete deterministic ML tie class obeys

$$\operatorname{diam}(\mathcal{J}_{\text{ML}}) \le \frac{A_{\text{tie}}}{Z}\sqrt{\frac{1-p}{p}} \le \frac{1-Z}{Z}\sqrt{\frac{1-p}{p}}. \tag{100}$$

Proof. The first inequality in (98) follows because two ML rules can differ only at an ambiguous syndrome. At each such syndrome, at least one unselected ML representative has the same probability as the selected one. These unselected representatives are decoding failures, so their total mass is at most $1 - Z$. This proves the second inequality.

Suppose first that $\mathcal{E}_D \ge \mathcal{E}_{D'}$ and put $\Delta_+ = T_D \setminus T_{D'}$. Equal ML weights give $\pi_p(\Delta_+) = \alpha(D, D')$. Every edge counted by T_D but not by $T_{D'}$ has an endpoint in Δ_+. Charge such an edge to one such endpoint. If $\operatorname{wt}(e) = j$, the total geometric-amplitude mass of all m incident edges is

$$\pi_p(e)\left(\frac{j}{\sqrt{q}} + (m-j)\sqrt{q}\right) \le \frac{m\pi_p(e)}{\sqrt{q}}, \tag{101}$$

where $0 < q < 1$. Summing (101) over Δ_+ and dividing by mZ proves (99). If the ordering of the two edge masses is reversed, exchange D and D'. Maximization over D, D' gives (100). □

In particular, for fixed $p \in (0, 1/2)$, if $A_{\text{tie}} = o(Z)$, every deterministic ML tie rule has the same asymptotic DQI score. The stronger, simpler condition $Z \to 1$ also forces the diameter to zero. The reliable quotient lift avoids this no-gain regime because its common acceptance tends to $1 - p$, not to one, and its controlled fiber tie has constant probability mass.

A. A secondary weight-enumerator estimate

The following classical estimate relates ambiguity to the code spectrum. If $c \in C \setminus \{0\}$ has weight d, then e and $e + c$ have equal weight only when d is even and e has $d/2$ ones on $\operatorname{supp}(c)$. Under $\text{BSC}(p)$ this event has probability

$$B_d(p) = \mathbf{1}\{d \text{ even}\}\binom{d}{d/2}[p(1-p)]^{d/2}, \tag{102}$$

$$B_d(p) \le \theta(p)^d, \qquad \theta(p) = 2\sqrt{p(1-p)} < 1. \tag{103}$$

A union bound yields

$$\alpha(D, D') \le \sum_{c\in C\setminus\{0\}} B_{\operatorname{wt}(c)}(p) \le \sum_{c\in C\setminus\{0\}} \theta(p)^{\operatorname{wt}(c)}. \tag{104}$$

This is a sufficient classical envelope, not the principal novelty. It can be looser than (98), especially for reliable decoders. For an $[m, k, d_{\min}]$ family, the simple condition $2^k\theta(p)^{d_{\min}} \to 0$ makes the right-hand side vanish.

VIII. A Secondary DE-NFG Evaluator and the Role of Bethe Analysis

The pair functional is naturally a two-copy contraction. This section makes that statement exact and separates it from any approximation claim.

For a deterministic decoder D, define the unnormalized accepted vector

$$|v_D\rangle = \sum_{e\in\mathbb{F}_2^m} p^{\operatorname{wt}(e)/2}(1-p)^{(m-\operatorname{wt}(e))/2} A_D(e)|e\rangle. \tag{105}$$

Let $E_{D,i}$ denote the undirected geometric-amplitude mass of accepted hypercube edges in coordinate i, so that $\sum_i E_{D,i} = mR_D$.

Theorem 15 (Deterministic decoder circuit to a strict DE-NFG)**.** *Suppose D is represented by a finite deterministic circuit with a unique internal-wire assignment for every input. Double every wire and replace a local Boolean indicator $\phi_g(u)$ by*

$$\Phi_g(u, u') = \phi_g(u)\phi_g(u'). \tag{106}$$

The resulting strict DE-NFG has closed contractions

$$Z_D = \langle v_D|v_D\rangle, \tag{107}$$

$$C_{D,i} = \langle v_D|(I + X_i)|v_D\rangle = Z_D + 2E_{D,i}. \tag{108}$$

Consequently,

$$R_D = \frac{1}{2m}\sum_{i=1}^{m}(C_{D,i} - Z_D). \tag{109}$$

Every doubled deterministic local factor in (106) *is rank-one positive semidefinite. The Bernoulli boundary is an outer product, and the inserted boundary matrix $I + X_i$ is real, entrywise nonnegative, and positive semidefinite.*

Proof. Expanding the doubled deterministic circuit produces $A_D(e)A_D(e')$ with the two Bernoulli amplitudes in (105). Closing corresponding wires with the identity sets $e' = e$ and gives (107). The X_i term sets $e' = e + \mathbf{e}_i$. Both orientations of every undirected accepted edge are present, which gives the factor two in (108). Summing the coordinate identities yields (109). Finally, $\Phi_g = \phi_g\phi_g^T$ and the Bernoulli boundary are outer products, while $I + X_i = \left[\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right]$ on bit i. These observations prove the stated positivity properties. □

The phase-averaged binary compilation is therefore real. Complex messages carry no additional exact information for this contraction. Before averaging a fixed binary instance, signed real factors can remain; over nonbinary alphabets, group characters can be genuinely complex. Neither case implies a decoding gain without a uniform algorithm and matched computational cost.

A. Exact and approximate inference boundaries

If the compiled normal factor graph is acyclic, one inward and one outward SPA sweep evaluates the relevant contractions exactly [15]. If its treewidth is w and the original wire alphabet has size d, exact elimination is polynomial in network size and exponential in the doubled width, with a representative factor $d^{2(w+1)}$. A private auxiliary tree whose exact contraction equals one original factor merely changes the elimination order. After a complete internal sweep it produces the same boundary update, the same partition function, and the same decoder. Such a refinement cannot create a strict gain by itself.

On a loopy compilation, the Bethe partition function and its stationary points supply approximations to the exact contractions. Every factor in the phase-averaged construction above is entrywise nonnegative. Treating each paired variable as one enlarged-alphabet variable therefore makes it an ordinary nonnegative factor graph, to which the standard finite-cover characterization applies [16]. No general complex DE-NFG cover theorem is needed here. For genuinely complex DE-NFGs, the analogous theorem is proved under a stated sufficient condition, while the unrestricted statement remains a conjecture [19]. We use the present network as an exact offline evaluator on trees or bounded-width graphs and treat loopy Bethe quantities as candidate surrogates. A surrogate earns an algorithmic claim only after it predicts decoder rankings more accurately or more cheaply than exact paired sampling in (15).

B. When one copy is enough

The universal optimizer in Theorem 8 satisfies $P(q) = W'_{\mathrm{ML}}(q)$. Hence a single-copy partition function and its derivative determine the optimal ML pair term; no doubled graph is needed if W_{ML} is already available. The DE-NFG remains useful for a restricted or non-ML decoder, whose exact loss is the two-point downward defect in (54). Paired sampling provides the direct statistical baseline in either case.

IX. Algorithms

The theorems yield exact targets and compact algorithms at different levels of structure.

A. Canonical ML section

For unequal bit probabilities, compare candidates by the ordered metric $(L, -G_\gamma, \text{monomial-order key})$ from (19); retain the first word with each new syndrome. When the probabilities and coordinate rewards are both uniform, this reduces to Hamming weight followed by the fixed monomial order. The resulting table is the globally DQI-optimal ML section from Theorems 3 and 8. An exact ML trellis or junction-tree solver can carry $-G_\gamma$ and the word key as secondary metrics, and an ML list decoder can apply the same rule to its minimum-likelihood candidates. The full table can be exponential in the codimension, so this is not a general polynomial-time ML construction. For $C_{d,n}$ in Theorem 12, the same canonical rule reduces to four weight comparisons and a deterministic tie test, hence runs in $O(N)$ time.

B. Aligned quotient-lift decoder

Given a base decoder D_0, the optimal member of the fixed-base fiber-lift ML class is the aligned rule in (69). It preserves the base representative weight syndrome by syndrome. Its cost is one base decoder call plus $O(n)$ bit operations, and its reversible implementation uses the same asymptotic work. The competing parity field in (70) also costs $O(n)$; hence the separation in Section VI is matched in decoding order and pointwise ML likelihood.

C. Restricted optimization on an explicit candidate graph

When an architecture restricts the available candidates, construct the pairwise compatibility rewards in (57). On a tree, a max-sum message from syndrome s to t is

$$M_{s\to t}(b) = \sum_{u\in N(s)\setminus t} \max_{a\in\mathcal{L}_u}\{M_{u\to s}(a) + \psi_{us}(a,b)\}, \quad (110)$$

where $b \in \mathcal{L}_s$ and ψ_{us} is the weighted one-bit compatibility indicator. Backtracking returns the exact section. A junction tree gives the bounded-treewidth complexities stated after Theorem 9. For two-label full matchings, offset propagation solves the zero-frustration case. For the quotient lift, the offsets vanish and the aligned field is immediate.

D. Stochastic selection among compact decoder architectures

When the full candidate graph is exponential, restrict attention to a compact family $\{D_\vartheta\}$, such as equivalent parity-check representations, redundant-check choices, message schedules, or damping parameters. If all candidates output minimum-weight representatives, Z is exactly common and the design objective is simply R_{D_ϑ}. Proposition 2 supplies a direct unbiased estimator:

1) sample a coordinate and the remaining Bernoulli error bits;
2) run D_ϑ on the two forced-bit syndromes;
3) record one if both returned representatives equal their inputs;
4) average and multiply by $\sqrt{p(1-p)}$.

The same random pairs can evaluate all candidates, reducing comparison variance. If the candidate family allows different acceptance, estimate Z separately and report both $\mathcal{J}$ and R; optimizing a conditional score alone can conceal an exponential retry penalty. A Bethe or DE-NFG surrogate is useful only in a regime where exact pair sampling is too costly and the surrogate's ranking is validated at matched compute.

X. Reproducible Numerical Audit

The numerical study certifies the asymmetric and radial optima, the reliability–sensitivity identity, the robust strict-gain family, and the quotient-lift identities; it also tests which broader algorithmic claims are supported. All reported decoders are deterministic and classical. The main experiments use real arithmetic; no BP with quantum messages is included. Machine-readable outputs, deterministic scripts, and a compact command list accompany this draft.

A. Asymmetric-product certificate

An exact-rational program tested Theorem 3 on 20 seeded random codes of lengths five through seven and five unequal log-likelihood patterns, each paired with unequal positive coordinate rewards. It chose $r_i = 2^{w_i}$ and $p_i = 1/(1 + 4^{w_i})$, so every probability and edge mass was rational. Every deterministic ML section was enumerated, with as many as 384 sections in one test. Every case passed downward closure of the canonical rule, the edge-charge identity (21), the two-term defect (22), and global optimality.

B. Reliability–sensitivity certificate

An exact-rational implementation independently computed Z_{ML}, its active one-sided directional derivative, the optimal accepted-edge mass, and the radial derivative and area identities. It passed 237 radial identities and 79 asymmetric directional identities. A separate adversarial enumerator checked 4,974 syndrome maps and 96,704 asymmetric ML sections, including rank-deficient representations, zero columns, repeated columns, and parameters near both open endpoints. It also found explicit counterexamples when the assumptions $p_i < 1/2$ or $\gamma_i > 0$ are removed.

C. Coefficientwise optimum certificate

An independent exhaustive program enumerated every deterministic ML section for the $[5, 2]$ code in (111) and for 16 seeded random full- rank parity-check matrices of lengths five through seven. Individual tests contained up to 3072 ML sections. In every case, it verified $N_k(T) \le kn_k$ coefficient by coefficient, the exact layer-defect identity, downward closure of the graded-lexicographic section, and $N_k(T_{\mathrm{glex}}) = kn_k$ for all k. At a positive test value of q, the enumerated maximizers were exactly the downward-closed ML sections.

D. Linear-source threshold certificate

An independent exact-arithmetic program checked the finite bridge used in Theorem 7. On 312 seeded full-rank maps of lengths four through ten, it verified transversal cardinality, downward closure, the accepted-edge/conditional-weight identity, and the top-mass converse. It also exhaustively enumerated every binary matrix in 104 small collision tests and verified the random-syndrome union bound, including rank-deficient maps. Exact rational checks confirmed the exponent derivative and DQI-deficit algebra above threshold.

E. Indecomposable-family certificate

Exact enumeration checked Theorem 12 for $d \in \{4, 6\}$ and three odd outer lengths beginning at $d - 1$, at $p = 0.24$. It verified the full coset partition, pointwise ML equality, minimum distance, rank, distinct columns, Tanner connectedness, indecomposability, and (82); the largest displayed edge-identity residual was 3.1×10^{-14}. At $n = 401$, the $d = 4$ conditional gap was 0.049372 against the limit 0.049865, and the acceptance-weighted per-attempt gap was 0.042206 against 0.042627. The corresponding $d = 6$ values were 0.028161 against 0.028582 and 0.025535 against 0.025917.

F. Positive-rate feasibility certificate

An independent exact certificate pairs the inner Rep_4 gadget with the self-dual $[8, 4, 4]$ $\mathrm{RM}(1, 3)$ outer code. It exhaustively verifies pointwise exact-ML equality, the coefficientwise version of (94), minimum distance four, nonzero distinct columns, and a connected row-equivalent check representation. A separate exponent audit evaluates (91) and (92) for a representative parameter choice and checks that the logarithmic-block union bound decreases to zero. These computations test the finite algebra and constants; the asymptotic existence statement in Theorem 13 follows from its analytic proof.

G. Quotient-lift certificate

For the lifted repetition family, exact enumeration was performed through full length $m = 20$. The tests checked syndrome consistency, equality of the selected weights, global ML optimality within every full coset, the matrix identity $H = H_0 Q$, connectedness of the Tanner graph, and (66) for five distinct tie fields. Every count of a failed identity was zero. The largest score-versus-cut residual was 9.98×10^{-17}.

Growing-length evaluation used $m \in \{8, 12, 20, 40, 80, 120, 200\}$. Table I reports the common acceptance and the separation between the constant and parity fields at $m = 200$. The predicted limits are (72) and (73). Convergence slows as $p \uparrow 1/2$ because the repetition-code binomial tail decays more slowly.

Measured decoding time rose from approximately $0.66\,\mu$s at $m = 8$ to $18.3\,\mu$s at $m = 200$. Linear fits for the aligned and parity rules both had $R^2 > 0.9995$. These timings are supporting checks, not hardware-independent complexity claims; the explicit bit-operation count is at most linear in m in addition to the base decoder.

H. Finite-code optimization beyond ML representatives

The main theorem compares ML tie fields. A separate exhaustive experiment asked whether allowing non-ML representatives can improve the acceptance-weighted per-attempt objective at a controlled acceptance loss. For the $[5, 2]$ code

$$C = \mathrm{span}\{00111, 11100\} \tag{111}$$

at $p = 0.40$, all $4^8 = 65{,}536$ transversals were enumerated. Table II gives the exact optima under two acceptance constraints. The finite-code improvement is genuine because R increases. It is not a growing-family theorem.

TABLE I
QUOTIENT-LIFT VALIDATION AT FULL LENGTH $m = 200$. THE TWO DECODERS ARE EXACT ML, POINTWISE EQUAL IN LIKELIHOOD, AND HAVE THE DISPLAYED COMMON ACCEPTANCE.

p	Common Z	$\Delta\mathcal{J}$	Limit of $\Delta\mathcal{J}$	ΔR	Limit of ΔR
0.05	0.950000	0.010788	0.010897	0.010249	0.010352
0.15	0.850000	0.053025	0.053561	0.045071	0.045527
0.25	0.750000	0.107171	0.108253	0.080378	0.081190
0.35	0.649995	0.165269	0.166939	0.107424	0.108511
0.45	0.506887	0.215787	0.223872	0.109379	0.123130

TABLE II
EXACT $[5, 2]$ TRANSVERSAL AUDIT AT $p = 0.40$.

Decoder constraint	Z	$\mathcal{J}$	R
ML/coset leader	0.406080	0.739737	0.097353
$Z \geq 0.95 Z_{\mathrm{ML}}$	0.388800	0.761279	0.101585
$Z \geq 0.80 Z_{\mathrm{ML}}$	0.360000	0.793939	0.105818

TABLE III
MATCHED-ACCEPTANCE FACTOR-GRAPH SELECTION.

m	Codes	Gains	Mean $\Delta\mathcal{J}$	Mean ΔR
20	12	12	5.88×10^{-4}	3.63×10^{-4}
40	5	5	3.19×10^{-4}	1.84×10^{-4}
100	3	3	5.6×10^{-5}	2.9×10^{-5}

Coordinate ascent and zero-temperature Bethe/max-sum followed by coordinate polishing reached the exact optimum in this small candidate space. A broader scan near the low-noise decoding threshold was negative: across sparse random matrices of lengths $10, 12, 16, 20$, no positive pair-mass gain over the compressed ML leader was found within the tested top-six candidate lists and acceptance fractions. The contrast prevents the five-bit example from being presented as an asymptotic phenomenon.

I. Equivalent-factor-graph selection at fixed ML acceptance

A practical compact candidate family varied invertible row representations, a small set of redundant checks, input scaling, and damping. Each output was projected to a stored minimum-weight tie, so all compared rules had identical ML acceptance. The total edge-update budget was fixed, and the baseline was the best tuned real BP rule on the identity representation. Offline selection used the exact pair metric. Results are summarized in Table III.

The canonical section in Theorem 8 supplies the exact ML ceiling for these tests: graph selection can move an approximate decoder toward that ceiling, but cannot exceed it. The observed gains decrease with length in this random family. At $m = 40$ and $m = 100$, some ML tie lists were truncated, so the results are lower bounds within the stored candidate pool rather than global tie optima. Direct pair-objective coordinate ascent did not improve the compressed lexicographic ML leader on the tested Hamming, BCH, cycle, and random-LDPC instances, as the theorem now predicts. The evidence supports finite-length graph and schedule selection for approximate decoding, not a nonvanishing random-LDPC claim.

J. Negative controls

Three controls delimit the interpretation.

1) A diagonal DE-NFG embedding reproduces ordinary real BP. A zero-phase rank-one amplitude parameterization reduces to a real prior-temperature rescaling. Neither produced a complex-specific gain against strengthened real baselines.
2) For a fixed binary DQI offset, the exact factors can be signed, but instance-tailored decoder selection was both slower and worse than directly optimizing the corresponding classical Max-XOR objective in the tested sizes. An unrestricted instance-aware decoder can encode a solution that was computed beforehand; its design cost cannot be omitted.
3) Loopy Bethe/max-sum was exact after polishing in the five-bit example, but its local ranking correlation varied by code and could be poor. Direct paired evaluation was cheaper whenever all syndromes could be enumerated.

These controls motivate the bounded DE-NFG claim in Section VIII: exact compilation is proved, while a general loopy approximation advantage remains open.

XI. DISCUSSION

A. What conventional decoder metrics miss

The quotient lift holds fixed every one-point statistic normally used to compare deterministic ML tie rules. The representative weight is identical for every syndrome, so the likelihood, frame-success probability, complete Hamming-layer profile, and expected retry count are identical. Nevertheless, the pair term changes by a weighted cut. The missing statistic is therefore not a finer shell average; it is adjacency among selected representatives. Theorem 8 sharpens this distinction: the common shell profile determines the largest attainable pair term, but it does not determine the value of an arbitrary tie rule. The downward defect measures exactly how far that rule lies below the shell-determined ceiling. This distinction complements recent work that sharpens shell-weighted DQI bounds [20]. A shell analysis and a pair analysis answer different questions.

B. When graph covers help

Two cover structures appear and need not coincide. First, an ML tie field can select a section of a covering graph

when its candidate compatibilities are full bijections. This is the decoder-design problem. Second, finite covers of a factor graph can characterize or approximate a Bethe partition function. This is an inference problem. The quotient lift solves the first problem exactly by gauge alignment. The DE-NFG compilation maps the DQI pair term to the second setting. Treating the two uses as distinct avoids the unsupported claim that a graph transformation automatically improves BP.

C. Implications for code design

The ambiguity bound identifies a tension. Large distance suppresses ML ties, which is favorable for ordinary reliable decoding but leaves little room for objective-aware tie-breaking. The quotient lift decouples the two roles: an arbitrary reliable base supplies error correction, while one weight-two fiber supplies controlled ambiguity. More general designs may distribute a small number of low-weight tie fibers through a reliable code and optimize their holonomy. The twisted family in Theorem 12 shows that a strict gap does not require weight-two codewords, repeated columns, or an algebraic direct product. Theorem 13 separately shows that positive rate, polynomial-time exact ML, nonvanishing acceptance, and a strict gap are mutually compatible, albeit in a logarithmic-block feasibility construction. Theorem 14 gives a necessary mass budget, and Theorem 8 gives the exact ML ceiling.

The open algorithmic problem is not to prove that BP is universally optimal. It is to realize the ordered rule $(L, -G_\gamma, \mathrm{key})$ compactly on useful code families when exact ML itself is costly. When both bit probabilities and coordinate rewards are uniform, this is minimum weight followed by the fixed monomial order. Trellis, junction-tree, and list decoders can carry these secondary keys exactly. For loopy approximate decoders, equivalent factor graphs and schedules can be ranked against the canonical ceiling. A DE-NFG Bethe approximation remains an offline surrogate, with Proposition 2 as the direct statistical baseline.

D. Quantum-algorithm boundary

The paper improves one component of a DQI implementation: the coherent effect of a classical syndrome decoder. The result establishes a strict state-quality separation at matched classical decoding success and cost. It does not by itself establish quantum supremacy, an end-to-end speedup, or an advantage on generic Max-XOR instances. Those claims additionally require an instance family, a complete reversible resource analysis, and comparison with the best classical optimizer. Extensions of DQI to weighted objectives and other association schemes broaden the possible application domain [22], [24], while known structural obstructions reinforce the need for instance-specific evidence [21].

XII. Conclusion

Maximum-likelihood ties are classically invisible to frame success but can be visible to the phase-averaged binary DQI accepted-pair objective. For independent probabilities $0 < p_i < 1/2$, positive coordinate rewards, and deterministic exact-ML sections, the ordered rule based on L, G_γ, and a multiplicative monomial order is globally optimal. Its exact defect splits into a fiberwise loss and a missing-boundary loss. More fundamentally, the optimal DQI interference is an exact directional sensitivity of the classical ML success probability; in the radial setting, the resulting area law links the entire DQI-deficit curve to the normalized log reliability. Applied to an existential family of full-rank linear syndrome maps at each fixed rate, it gives an explicit Shannon threshold, correct-decoding exponent, and limiting optimal DQI score. A canonical section simultaneously maximizes every radial layer coefficient. The selected ambiguous mass gives a quantitative no-gain bound, while explicit families show a strict constant separation when that budget persists. One family is connected, coordinate-indecomposable, and linear-time; a complementary family has positive rate and polynomial-time exact ML. The DE-NFG construction remains an offline realization of the pair functional, not an automatic decoding gain. Together, these results give an exact optimizer, a reliability–interference identity, and matched sharpness and stability regimes for objective-aware decoder co-design.

Appendix A
The Uniform-Prior Endpoint

At $p = 1/2$ with uniform coordinate rewards $\gamma_i = 1$, every transversal has 2^ρ words and common acceptance $2^{\rho-m}$. The DQI optimization becomes an induced-edge problem in the hypercube. Nonuniform rewards require a separate weighted endpoint problem and are not covered by this proposition.

Proposition 16 (Exact optimum at the uniform endpoint)**.** *For every binary linear code with redundancy ρ,*

$$P_D(1) \le \rho 2^{\rho-1}. \tag{112}$$

Choose any ρ linearly independent columns of H. The coordinate ρ-subcube supported on those columns is a transversal and attains equality. Hence

$$\max_D \mathcal{J}_D = \frac{1}{2} + \frac{\rho}{2m} \qquad (p = 1/2). \tag{113}$$

Proof. The edge-isoperimetric inequality for the m-cube states that a set A induces at most $|A| \log_2 |A|/2$ edges [25]. Substituting $|A| = 2^\rho$ gives (112). If I indexes ρ independent columns, the restriction of H to the coordinate subspace on I is bijective onto $\mathbb{F}_2^\rho$. That subcube is therefore a transversal and attains the isoperimetric bound. Equation (113) follows from Theorem 1 by continuity at $q = 1$. □

When the code dimension grows linearly, the common acceptance $2^{-(m-\rho)}$ is exponentially small. Proposition 16 is an exact combinatorial endpoint, not an efficient postselection claim.

Appendix B
Fixed-Instance Signs and a No-Free-Lunch Boundary

Before averaging a known binary instance offset $v \in \mathbb{F}_2^m$, a syndrome-edge term can carry a sign. With a reference section D^0 and labels $c_s \in C$, the sign on the edge $t = s + h_i$ is

$$(-1)^{v\cdot(D^0(s)+e_i+D^0(t)+c_s+c_t)}. \tag{114}$$

The label-dependent terms are vertex gauges. Around a closed syndrome walk, the gauge-invariant product is $(-1)^{v\cdot c_\gamma}$, where $c_\gamma \in C$ is the physical codeword traced by the walk. All cycle signs can be gauged positive exactly when $v \in C^\perp = \operatorname{row} H$. Thus binary fixed-instance models require signed real factors; complex characters become intrinsic only over larger alphabets.

There is also an algorithmic quantifier trap. At the uniform point, consider a linear section $D_G(s) = Gs$ with $HG = I_\rho$. Its Walsh output is the basis label

$$x = G^T v. \tag{115}$$

Proposition 17 (An instance-aware right inverse can hard-code an answer)**.** *If $v \notin \operatorname{row} H$, then for every target $x \in \mathbb{F}_2^\rho$ there exists a right inverse G with $G^T v = x$. If $v = H^T a$ lies in $\operatorname{row} H$, every right inverse satisfies $G^T v = a$.*

Proof. If $v \notin \operatorname{row} H$, the augmented matrix $\left[\begin{smallmatrix} H \\ v^T \end{smallmatrix}\right]$ has rank $\rho + 1$. For each column j, solve

$$\begin{bmatrix} H \\ v^T \end{bmatrix} g_j = \begin{bmatrix} e_j \\ x_j \end{bmatrix}. \tag{116}$$

The resulting G obeys both required equations. If $v = H^T a$, then $G^T v = G^T H^T a = (HG)^T a = a$. □

Choosing $G(v)$ to output an optimizer first requires knowing that optimizer; the linear system then only encodes it in the decoder. Meaningful fixed-instance co-design therefore uses one uniform compact rule for all v or charges the full cost of constructing the instance-dependent decoder.

Appendix C
Reproducibility Map

The accompanying research archive separates theorem certificates from exploratory searches.

TABLE IV
Primary reproducibility artifacts.

Artifact	Purpose
`asymmetric_ml_downset_certificate.py`	Exact product-prior optimizer and two-term defect checks.
`dqi_reliability_sensitivity_audit.py`	Exact directional-derivative and area identities.
`linear_source_dqi_threshold_certificate.py`	Finite score bridge, converse, collision bound, and threshold algebra.
`adversarial_theorem_audit.py`	Independent scope and counterexample search.
`ml_downset_optimum_certificate.py`	Exhaustive coefficientwise ML optimum and defect checks.
`twisted_repetition_ambiguity_certificate.py`	Exact robust-family ML, distance, graph, and gap checks.
`positive_rate_strict_gain_certificate.py`	Exact positive-rate product identity and logarithmic-block exponent audit.
`dqi_quotient_lift_validation.py`	Independent quotient, ML, cut, asymptotic, and timing checks.
`ml_tie_lift_certificate.py`	Symbolic and growing-length certificate for the reliable lift.
`transversal_exact_certificate.py`	Exhaustive $[5, 2]$ transversal envelope.
`dqi_ml_tie_codesign.py`	Equivalent-factor-graph and matched-acceptance tie selection.
`dqi_pair_sampling.py`	Direct paired Monte Carlo identity and concentration checks.
`fixed_v_right_inverse_certificate.py`	Exhaustive check of Proposition 17.

The generated CSV, JSON, and Markdown summaries record parameter grids, finite-length identities, and random seeds. The principal theorem does not depend on floating-point evidence; the numerics independently test its implementation and asymptotic interpretation.

References

[1] S. P. Jordan, N. Shutty, M. Wootters, *et al.*, "Optimization by decoded quantum interferometry," *Nature*, vol. 646, no. 8086, pp. 831–836, Oct. 2025, doi: 10.1038/s41586-025-09527-5.

[2] L.-H. Chang, P.-N. Chen, F. Alajaji, and Y. S. Han, "Decoder ties do not affect the error exponent of the memoryless binary symmetric channel," *IEEE Trans. Inf. Theory*, vol. 68, no. 6, pp. 3501–3510, Jun. 2022, doi: 10.1109/TIT.2022.3150597.

[3] L.-H. Chang, P.-N. Chen, and F. Alajaji, "On decoder ties for the binary symmetric channel with arbitrarily distributed input," *Entropy*, vol. 25, no. 4, Art. no. 668, Apr. 2023, doi: 10.3390/e25040668.

[4] A. B. Kiely, J. T. Coffey, and M. R. Bell, "Optimal information bit decoding of linear block codes," *IEEE Trans. Inf. Theory*, vol. 41, no. 1, pp. 130–140, Jan. 1995, doi: 10.1109/18.370113.

[5] G. A. Margulis, "Probabilistic characteristics of graphs with large connectivity," *Problems Inf. Transm.*, vol. 10, no. 2, pp. 174–179, 1974.

[6] L. Russo, "On the critical percolation probabilities," *Z. Wahrscheinlichkeitstheorie Verw. Gebiete*, vol. 56, no. 2, pp. 229–237, 1981, doi: 10.1007/BF00535742.

[7] C. Méasson, A. Montanari, T. J. Richardson, and R. L. Urbanke, "The generalized area theorem and some of its consequences," *IEEE Trans. Inf. Theory*, vol. 55, no. 11, pp. 4793–4821, Nov. 2009, doi: 10.1109/TIT.2009.2030457.

[8] J. Chen, D.-K. He, A. Jagmohan, L. A. Lastras-Montaño, and E.-H. Yang, "On the linear codebook-level duality between Slepian–Wolf coding and channel coding," *IEEE Trans. Inf. Theory*, vol. 55, no. 12, pp. 5575–5590, Dec. 2009, doi: 10.1109/TIT.2009.2032815.

[9] T. Helleseth, T. Kløve, and V. I. Levenshtein, "Error-correction capability of binary linear codes," *IEEE Trans. Inf. Theory*, vol. 51, no. 4, pp. 1408–1423, Apr. 2005, doi: 10.1109/TIT.2005.844080.

[10] K. Yasunaga and T. Fujiwara, "On correctable errors of binary linear codes," *IEEE Trans. Inf. Theory*, vol. 56, no. 6, pp. 2537–2548, Jun. 2010, doi: 10.1109/TIT.2010.2046209.

[11] S. Bitzer, A. Ravagnani, and V. Weger, "Weighted-Hamming metric for parallel channels," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Athens, Greece, 2024, pp. 2616–2621, doi: 10.1109/ISIT57864.2024.10619239.

[12] M. Borges-Quintana, M. A. Borges-Trenard, and E. Martínez-Moro, "Computing coset leaders of binary codes," arXiv:1001.5074, Jan. 2010.

[13] M. Borges-Quintana, M. A. Borges-Trenard, I. Márquez-Corbella, and E. Martínez-Moro, "Computing coset leaders and leader codewords of binary codes," *J. Algebra Appl.*, vol. 14, no. 8, Art. no. 1550128, 2015, doi: 10.1142/S0219498815501285.

[14] J. A. Grochow and J. Tucker-Foltz, "Computational topology and the Unique Games Conjecture," in *Proc. 34th Int. Symp. Comput. Geom. (SoCG)*, vol. 99, 2018, pp. 43:1–43:16, doi: 10.4230/LIPIcs.SoCG.2018.43.

[15] F. R. Kschischang, B. J. Frey, and H.-A. Loeliger, "Factor graphs and the sum-product algorithm," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 498–519, Feb. 2001, doi: 10.1109/18.910572.

[16] P. O. Vontobel, "Counting in graph covers: A combinatorial characterization of the Bethe entropy function," *IEEE Trans. Inf. Theory*, vol. 59, no. 9, pp. 6018–6048, Sep. 2013, doi: 10.1109/TIT.2013.2264715.

[17] H.-A. Loeliger and P. O. Vontobel, "Factor graphs for quantum probabilities," *IEEE Trans. Inf. Theory*, vol. 63, no. 9, pp. 5642–5665, Sep. 2017, doi: 10.1109/TIT.2017.2716422.

[18] M. X. Cao and P. O. Vontobel, "Double-edge factor graphs: Definition, properties, and examples," in *Proc. IEEE Inf. Theory Workshop (ITW)*, Kaohsiung, Taiwan, 2017, pp. 136–140, doi: 10.1109/ITW.2017.8277985.

[19] Y. Huang and P. O. Vontobel, "Graph-cover-based characterization of the Bethe partition function of double-edge factor graphs," arXiv:2506.16250, Jun. 2025.

[20] M. Gao and Y. Chang, "Hidden quantum advantage near the decoding threshold of decoded quantum interferometry," arXiv:2604.15025, Apr. 2026.

[21] E. R. Anschuetz, D. Gamarnik, and J. Z. Lu, "Spin glass transitions obstruct decoded quantum interferometry," arXiv:2509.14509v2, Aug. 2026.

[22] K. Bu, W. Gu, and X. Li, "Multivariate decoded quantum interferometry for weighted optimization," arXiv:2605.10666v2, May 2026.
[23] M. J. Kramer, C. Schubert, and J. Eisert, "Approximability limits for bounded-degree max-LINSAT and implications for decoded quantum interferometry," arXiv:2606.13570, Jun. 2026.
[24] A. Krajenbrink, C. Krawchuk, A. Rosmanis, and M. Rosenkranz, "Decoded quantum interferometry beyond Hamming: Rank-metric and translation association schemes," arXiv:2606.04843, Jun. 2026.
[25] L. H. Harper, "Optimal numberings and isoperimetric problems on graphs," *J. Combin. Theory*, vol. 1, no. 3, pp. 385–393, 1966, doi: 10.1016/S0021-9800(66)80059-5.